ALICE L WALTON SCHOOL OF MEDICINE

β–Blockers

Last updated November 4, 2025 12 min read ScholarRx

Adapted by: Diptiman Bose, Trager Hintze

Editors: Jordan Finn, Russell Johnson

Adapted from: β-Blockers

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action of beta-adrenergic blockers.
2. Compare the selectivity of beta-adrenergic blockers and predict their uses in pharmacotherapy.
3. Differentiate the pharmacokinetic properties of beta-adrenergic blockers and explain their clinical implications.
4. Discuss the potential adverse effects and contraindications associated with beta-adrenergic blocker use.

CASE CONNECTION

YW is a 66-year-old female with type 2 diabetes, hypertension, and hypothyroidism. She is opposed to using injectable medications, and, because of financial constraints, the only medications she can use are metformin and a sulfonylurea. Her blood glucose is fairly well controlled. She takes three medications for her blood pressure: an ACE inhibitor, a diuretic, and a calcium-channel blocker. However, her systolic pressures have been in the 160s, and another medication is needed. She is not taking a β-blocker.

What are the benefits and disadvantages of β-blockade for YW? Consider your answer as you read, and we'll revisit YW at the end of the brick.

GO TO CONCLUSION ↓

What Are β–Blockers?

β-Blockers are a prominent heterogeneous class of medications with pharmacological characteristics that make them useful in treating various cardiovascular diseases. β-Blockers, also known as β-adrenoceptor blockers, are a class of drugs that block β-adrenergic receptors. They are also known as sympatholytic drugs (sympatho= SNS, lytic=breaking down), which are agents that inhibit the effect of the sympathetic pathway of the autonomic nervous system. Hence, their [illegible]mimetic drugs, which "mimic" or duplicate the sympathomimetic effects. Each type of β-blocker is associated with particular receptors and physiologic processes, usually related to the heart and blood vessels. If you recall from the ANS review, the β1-receptors play key role in the function and regulation of cardiac (heart) and renal (kidney) functions. β2: the heart, lungs, and the kidneys. β_3-receptors, which are found in the gallbladder, bladder, and brown fat and are not important clinically.

HOW DO β–BLOCKERS WORK? (LO–1)

As you recall, β-adrenergic receptor isoforms are expressed in various organ systems with different levels of isoform expression patterns. The heart, lungs, and kidneys were organs where β-receptor functions were predominant.

In the Heart, the β1-receptors control impulse conduction through the sinoatrial (SA) node and atrioventricular (AV) node, and agonism of β1-receptors increases the force of cardiomyocyte contraction (inotropy). Thus, inhibition of β1-receptors (or antagonism) in the heart decreases β1-mediated effects, such as control of heart rate (HR) and myocardial contractility. Since HR and myocardial contractility decrease, β1-antagonists exhibit negative chronotropic and negative inotropic effects (opposite effects from sympathomimetic catecholamine epinephrine). This decrease is most evident when demand on the heart is high, such as during stress or exercise, when β1-receptors produce a ceiling on maximal heart rate.

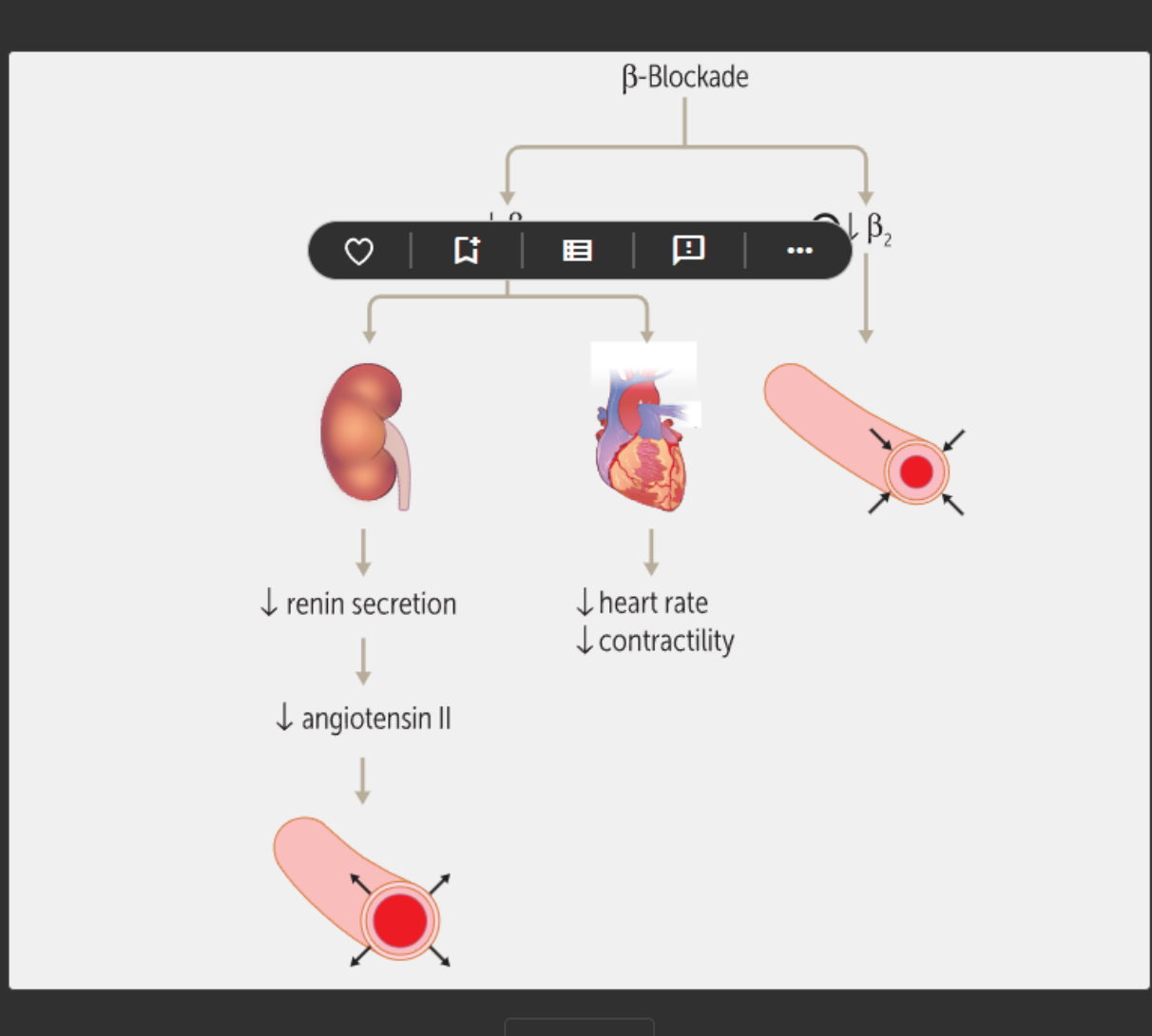
In the Kidneys, β1-receptors stimulate renin release, the first step in activating the renin-angiotensin-aldosterone system (RAAS). This causes water retention, which expands blood volume. It also causes angiotensin II-mediated vasoconstriction, thu[illegible] of β1-receptors inhibits these responses. This is the proposed mechanism of decreasing blood pressure using β-blockers.

In the lungs, β-blockers can inhibit β2-receptor function in the lungs, decreasing pulmonary function. In normal individuals, this effect may be minimal; however, in patients with COPD, asthma, or other respiratory dysfunctions, this β_2-receptor inhibition might be life-threatening. This is an undesirable effect of β-antagonism and we will discuss this further in the following sections.

CLINICAL CORRELATION

In systolic heart failure, the combined sympathetic effects lead to increased myocardial oxygen demands, which may result in pathologic remodeling of the heart. β-Blockers are used to treat systolic heart failure because they negate these harmful effects on the heart. β-Blockers should be avoided in patients with very low cardiac output, such as those experiencing cardiogenic shock.

Figure 1 summarizes the effects of β_1-receptor blockade in the heart and kidneys.



QUIZ

Figure 1

What is the effect of β_2-blockade in the lungs?

WHAT ARE THE TYPES OF β–blockers?(LO–2)

Because there are different categories of β-adrenergic receptors, each with distinct actions, we start by reviewing what each type does. This will be important in our clinical discussion: not all β-blockers inhibit all the receptor types. β-blockers are categorized by properties such as relative affinity for β1 and β2 receptors, intrinsic sympathomimetic activity, α-blockade, lipid solubility, vasodilation, and of course, pharmacokinetics. The easiest way is to group them by whether they are selective for β1-receptors, selective for β2-receptors, or nonselective (meaning that they bind to both receptor types). The easiest way to remember all β-blockers is that they all end **in -ol.** β-blockers based on their selectivity are shown in table 1 below.

Table 1

Classification	Drug Names
β_1-selective (cardioselective) blockers	Acebutolol,* atenolol, betaxolol, bisoprolol, celiprolol, esmolol, metoprolol, nebivolol, oxprenolol*
Nonselective β-blockers	Carteolol eye drops, timolol, nadolol, pindolol,* propranolol,
Nonselective β-a[illegible]	[illegible]l, labetalol*

Mnemonic

Some common β-blockers that are β_1 selective = **MAN BABE**
Metoprolol, **A**tenolol, **N**ebivolol,
Betaxolol, **A**cebutolol, **B**isoprolol, **E**smolol

What type of β-blocker is atenolol?

HOW DOES THE SELECTIVITY OF β–BLOCKERS HELP CLINICALLY? (LO–3)

Uses in Cardiovascular Disease

- *Myocardial infarction and Angina:* Cardioselective β-blockers are now first-line drugs in managing acute coronary syndrome (myocardial infarction and unstable angina). One of the principal causes of myocardial infarction is a mismatch in the supply and demand of oxygen to cardiac muscle, known as ischemic heart disease. An increased oxygen demand by the heart cannot be met because of coronary occlusions or vasospasm, which results in ischemia and eventual death of cardiac tissue. The **negative inotropic and chronotropic effects** of β-blockers can help alleviate the mismatch by decreasing the myocardial contractility and heart rate. Since our desired effect is β1 selective, in this case, a selective β-blocker such as Metoprolol is often used for treatment. While all β-blockers exhibit these effects, in this case the use of β_1-selective blockers such as Metoprolol makes more sense. After the resolution of a myocardial infarction (cardiac arrest), continued use of a β-blocker has been shown to be highly beneficial.
- *Congestive Heart Failure:* In congestive heart failure (CHF), the reduced ability of the heart to maintain cardiac output results in massive physiologic changes especially increased sympathetic nervous system activation. The increased SNS stress of a failing heart makes the disease worse. The addition of β-blockers in CHF treatment has been shown to improve mortality and morbidity. Note that the potentially adverse effects of decreased cardiac output from β_1-blockade are overcome by the positive effects. Specifically, three β-blockers, Metoprolol, Carvedilol, and Bisoprolol, are us[illegible]
- *Hypertension:* Despite their functional utility in reducing blood pressure over time, β-blockers are not first line for treating hypertension, even though they are involved in multiple pathways that control blood pressure, including the RAAS and cardiac output control. Notice that only their demonstrated β_1-mechanisms would lower blood pressure (by reducing cardiac output and reducing renin/angiotensin II). They don't affect the vasoconstriction of α_1-receptors, and β_2 blockade would vasoconstrict the arteries and thus increase the blood pressure.
- *Arrhythmias:* β-blockers are categorized as negative inotropic and chronotropic agents. Inhibition *of* β_1-receptors can slow down SA and AV nodal functions. Slowing down the SA and AV nodal function reduces the risk of arrhythmias. Hence, β-blockers are pharmacologically classified as anti-arrhythmics and used in the treatment of arrhythmia.

Uses in Non-Cardiovascular

- *Hyperthyroidism:* β-blockade can provide useful symptomatic treatment for patients by reducing sympathetic nervous system activation. Non-selective β-blockers such as Propronolol have been shown to useful.
- *Anxiety:* Inhibition of β_1-receptors with Propranolol can also help patients with performance anxiety and muscle tremors from SNS activitation. When used prophylactically, it can help with public speaking and anxiety-provoking scenarios.
- *Glaucoma:* Glaucoma Topical administration of β-blockers (Timolol, Carteolol?) has been shown to decrease aqueous humor production.
- *Migraine:* Although the mechanism is unknown, β-blockers such as Propranolol and Metoprolol have been shown to beneficial in the prophylactic use

WHAT ARE THE IMPLICATIONS OF DIFFERENT PHARMACOKINETIC PROFILES AMONG Β–BLOCKERS? (LO–3)

While β-blockers share similar pharmacotherapeutic effects, their pharmacokinetic properties vary greatly, impacting their clinical effectiveness and potential side effects. While there are many factors that regulate effectiveness, we will focus on lipid solubility, metabolism, intrinsic sympathomimetic activity, and renal clearance for now. The following lists β-blockers with distinct PK/PD properties:

- *Propranolol*: Propranolol (non-selective) is the most lipophilic β-blocker. It has a high volume of distribution. Although the plasma half-life is short, it can cross the blood brain barrier (BBB) and elicit CNS effects.
- *Esmolol:* Esmolol is the β1-selective blocker metabolized by the blood esterases. Hence, it is formulated as an IV only. The onset of action is within 5-10 minutes of administration.
- *Acebutolol and Pindolol:* These β-blockers have intrinsic sympathomimetic activity (ISA), also called partial agonist activity. They provide low-grade beta stimulation at rest but act as typical beta blockers when sympathetic activity is high. These drugs are contraindicated in patients with angina because the ISA can exacerbate SNS tone/activity, which could precipitate anginal episodes.
- *Metoprolol:* Metoprolol is lipid-soluble (less than Propranolol), almost completely absorbed by the small intestine, and largely metabolized by the liver. It is available as a succinate and tartrate salt and can be administered IV or orally (IR and ER). The succinate salt has a longer half-life than the tartrate.
- *Atenolol:* Atenolol is more hydrophilic and eliminated unchanged in the kidneys. It has longer half-life.
- *Labetalol:* Labetalol (non-selective β-blocker and α1-blocker) is metabolized extensively by the liver. It is available in oral and IV formulations. The α1 to β is about 1:3 (oral) and 1:7 (IV). The onset of action for IV is within 5 minutes, with peak effect in 5-15 minutes.

CLINICAL CORRELATION

Labetalol is a nonselective β-blocker that also has weak α_1-blocking effects. Because α_1-receptors cause vasoconstriction, it is a potent antihypertensive used intravenously and orally for a hypertensive emergency (ie, severe symptomatic hypertension).

WHAT ADVERSE DRUG REACTIONS (ADRS) ARE ASSOCIATED WITH Β–BLOCKER USE? (LO–4)

- ***Cardiovascular effects:*** The same effects on heart rate and contractility that are advantageous in myocardial infarction and heart failure can also be deleterious. If a patient already has bradycardia, a decreased heart rate can be dangerous, especially in patients with AV conduction defects. It's also important to note that acute withdrawal of β-blockers can result in exacerbation of angina. This maybe be due to the upregulation of β_1-receptors and their increased sensitivity. Gradual tapering of the dose is generally recomm[illegible]
- ***Respiratory effects:*** β-blockers can cause bronchoconstriction with β2-blockade. β2-Receptors in the bronchioles mediate this effect, which can be especially problematic in patients who already have difficulty breathing, like those with chronic obstructive pulmonary disorder (COPD) or asthma. These patients can develop increased wheezing, hypoxia, and even respiratory failure during treatment with a β-blocker. β-blockers with β_1-selectivity are less likely to cause these effects.
- ***Metabolism:*** β_2-Blockade can result in hypoglycemia and mask the recognition of insulin-induced hypoglycemia. If a diabetic patient is prone to hypoglycemia, β-blockers should be used with caution. β_1-selective blockers maybe preferred. β-Blockade of the central and autonomic nervous systems causes adverse effects, including depression, fatigue, and sexual dysfunction, which are usually the major concerns of patients taking these medications. Hyperkalemia is uncommon but can be seen in patients with chronic kidney disease who are already prone to hyperkalemia.

What is the main adverse respiratory effect associated with noncardioselective β-blockers?

CLINICAL CORRELATION

In patients who present emergently with myocardial infarction secondary to acute cocaine intoxication, nonselective β-blockade is contraindicated. Vasodilation in the skeletal muscle arteries would be blocked as a result of β_2-receptor effects, which would result in unopposed α action. This could result in a hypertensive crisis, possibly further damaging cardiac tissue.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to YW, what are the benefits and disadvantages of β-blockade for her condition?

Because patients with diabetes have high rates of heart disease, it is beneficial to control heart rate and blood pressure. However, β-blockade can also mask the symptoms of hypoglycemia (a risk in patients taking sulfonylurea) or even affect blood sugar (and insulin resistance). If YW begins taking a β-blocker, you must pay careful attention to monitoring blood sugars and the associated symptoms, but, in YW's situation, the benefits likely outweigh the risks.

Summary

- β-Blockers are a class of drugs that inhibit the sympathetic pathway of the autonomic nervous system by blocking β-adrenergic receptors.
- They are categorized into nonselective (β1 and β2), cardioselective (β1 only), and combined α-β drugs.
- β1-receptors are primarily located in the heart and kidneys, controlling heart rate and contractility, and their blockade results in decreased heart rate and contractility.
- β2-receptors are primarily located in the lungs, skeletal muscle vasculature, and liver, and their blockade can lead to increased airway resistance and metabolic changes.
- β-Blockers are used to treat various cardiovascular diseases such as acute coronary syndrome, congestive heart failure, and atrial fibrillation, as well as noncardiovascular conditions such as tachycardia associated with hyperthyroidism and performance anxiety.
- Adverse effects of β-blockers include bronchoconstriction, decreased heart rate, decreased cardiac output, depression, and sexual dysfunction.

Review Questions

1. What is a primary effect of β1-receptor blockade in the heart?

A. ● Decrease heart rate
B. ○ Increase heart rate
C. ○ Increase blood pressure
D. ○ Increase [illegible]

Hide Explanation

The correct answer is decrease in heart rate (A). Blockade of β1-receptors reduces heart rate and contractility, leading to a drop in cardiac output. Increase in heart rate (B) is incorrect as β1-receptor blockade decreases heart rate. Increase in contractility (C) is incorrect because β1-receptor blockade reduces contractility. Increase in renin release (D) is incorrect as β1-receptor blockade inhibits renin release. Increase in blood pressure (E) is incorrect because β1-receptor blockade tends to lower blood pressure.

2. Which β-blocker is most lipophilic?

A. ● Propranolol
B. ○ Timolol
C. ○ Metoprolol
D. ○ Atenolol

Hide Explanation

The correct answer is Propranolol (A). Propranolol is the most lipophilic β-blocker, allowing it to cross the blood-brain barrier and elicit CNS effects. Esmolol (C) is β1-selective and metabolized by blood esterases is hydrophilic. Metoprolol (C) is lipid-soluble but less than [illegible]ic.

3. Which β1-selective blocker is used in treating CHF?

A. ○ Propranolol
B. ○ Labetalol
C. ○ Esmolol
D. ● Metoprolol

Hide Explanation

The correct answer is Metoprolol (D). Metoprolol is a β1-selective blocker, indicated for CHF. Propranolol (A) is non-selective, affecting both β1 and β2 receptors. Labetelol (B) is a mixed β- blocker and Esmolol is selective but not recommended for CHF treatment.

4. A non-selective b-blocker would be inappropriate for asthma patients because they are cause ___ .

A. ○ Hyperglycemia
B. ● Bronchoconstriction
C. ○ Hypokale[illegible]
D. ○ Tachycardia

Hide Explanation

The correct answer is bronchoconstriction (B). β2-blockade from non-selective b-block can cause bronchoconstriction, which is especially problematic for asthma patients. (A) Hyperglycemia is not a β-blocker ADR in patients with AV conduction defects. Hyperkalemia (C) occurs with β-blocker use. Tachycardia (D) is not seen, bradycardia is more common.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of beta-adrenergic blockers.~~

02 ☑ ~~Compare the selectivity of beta-adrenergic blockers and predict their uses in pharmacotherapy.~~

03 ☑ ~~Differen[illegible]ta-adrenergic blockers and explain their clinical implications.~~

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Objective Confidence: 4/4